Daniel Brady

Electrical Engineering | Atlanta, GA dcb2048@gmail.com

EDUCATION

Georgia Institute of Technology
Expected May 2028

B.S. in Electrical Engineering | GPA 4.00/4.00

ENGINEERING EXPERIENCE

AVL Technologies | Asheville, NC
Summer 2026

Software and Controls Intern

- Supported embedded controls for a three-axis satellite communications antenna. Traced PIC32 firmware interactions with Ethernet, motor control, and RF receiver interfaces.
- Built an ADC test project in MPLAB X and XC32 to verify channel conversion behavior and investigate device programming, linker, and connectivity issues.

Duke Energy | Raleigh, NC
Spring 2026

Electrical Engineering Co-op, Transmission Substation Engineering

- Revised five MicroStation drawing types for a 230 kV breaker replacement, including one-line and three-line diagrams, planning, plan and section views, and conduit and cable drawings.
- Incorporated feedback from electrical, mechanical, and civil teams and engineering technologists during design package review.

SELECTED PROJECTS

Current-Starved CMOS Ring Oscillator
Fall 2026

Silicon Jackets | Analog Design

- Designed and simulated a three-stage oscillator in Cadence Virtuoso targeting 500 MHz at a 1.8 V supply with a buffered 1 pF output load. Completed physical layout and passed DRC and LVS.

Handwritten Digit Classifier
Fall 2026

Georgia Tech | ECE 2036

- Implemented and tested a C++ neural network for MNIST digit classification. Compared accuracy and runtime across hidden-layer widths.

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB, R, LC-3 assembly

Tools: Cadence Virtuoso, MPLAB X, XC32, Bentley MicroStation, AutoCAD, Git, Wireshark

Hardware: CMOS schematic and layout, PIC32 microcontrollers, ADCs, Ethernet interfaces

INVOLVEMENT

Georgia Tech: Silicon Jackets, Swim Club, Society of Sales Engineering

Independent tutor: Calculus and physics